

Overview:

This report covers the development and findings of a market intelligence dashboard built to analyze the global uranium market. The project was built to demonstrate a complete BI workflow from start to finish: sourcing and cleaning raw public data, structuring it into a relational database, computing key performance indicators through SQL, and presenting everything through an interactive visual dashboard.

The uranium market made for a good case study. It is data-rich, economically significant, and involves the kind of multi-source data that analyst roles actually deal with. Canada is also one of the world's top producers, which gives the analysis some real regional relevance rather than being purely academic.

Objective:

The project set out to answer three straightforward business questions:

Where is the uranium spot price today, and how has it trended over time? Which countries dominate global production, and where does Canada rank? And what KPIs would an energy analyst actually want to see first thing in the morning?

Data and Methodology:

Three public datasets were sourced and cleaned.

The first was uranium spot pricing, monthly USD/lb figures from 1990 to 2026 sourced from Macrotrends. This gave over 35 years of price history, including the 2007 spike, the long decline after Fukushima, and the recent recovery.

The second was uranium production by country, covering annual output in tonnes from 2013 to 2024, sourced from the World Nuclear Association across the eight largest producing nations.

The third was nuclear electricity generation by country, annual TWh figures from 2013 to 2024 from Our World in Data, used to put production numbers in context relative to consumption.

All three datasets were cleaned and standardized using Python and Pandas, then loaded into a SQLite database built around a fact and dimension schema. Keeping measures like prices and production volumes separate from dimensions like country and time period made the KPI queries straightforward and the database easy to work with. Six SQL queries were written to calculate the KPIs, and the results feed directly into the HTML dashboard rendered with Chart.js.

Key Findings:

The uranium spot price has climbed significantly since 2021, recovering from a decade-long slump that followed the Fukushima disaster in 2011. As of May 2026, the price sits at \$88.30/lb, up 19.0% year-over-year and 2.6% month-over-month. The 12-month average of \$78.32/lb confirms this is a sustained trend rather than a short-term spike.

On the production side, the global market is heavily concentrated. Kazakhstan alone accounts for 43% of world supply. Canada sits second at 15.3%, with most of that coming from the Athabasca Basin in Saskatchewan, which holds some of the highest-grade deposits in the world. The top three producers together control the large majority of global output, meaning any supply disruption in one of them has real price consequences.

Canada's position is worth noting specifically. Its high-grade deposits give it a cost and quality edge over lower-grade operations in other countries, which is part of why it holds a strong second-place rank despite producing a fraction of what Kazakhstan does.

Dashboard Summary:

The dashboard runs across two pages. The Executive Overview shows the five KPI cards up front alongside a full price history chart going back to 1990 and a top producers bar chart. The second page goes deeper into trends, with a 12-month moving average overlaid on spot price, a world production trend line, a stacked country production chart, and a direct Canada versus Kazakhstan comparison.

The whole thing runs in the browser with no server required, which makes it easy to share and open anywhere.

Reflections:

The most useful part of this project was working out the database schema before touching any queries. Getting the structure right upfront meant the KPI queries came together quickly and cleanly without any messy workarounds. The other valuable piece was handling three datasets that each came in different formats, with different time granularities and inconsistent country naming conventions. Deciding how to handle those gaps and inconsistencies in the cleaning step was where most of the actual thinking happened.

More than anything, the project reinforced that good BI work depends just as much on data architecture and judgment calls as it does on building a nice-looking dashboard at the end.

Data sources: UxC / Macrotrends, World Nuclear Association, Our World in Data. All figures as of May 2026.